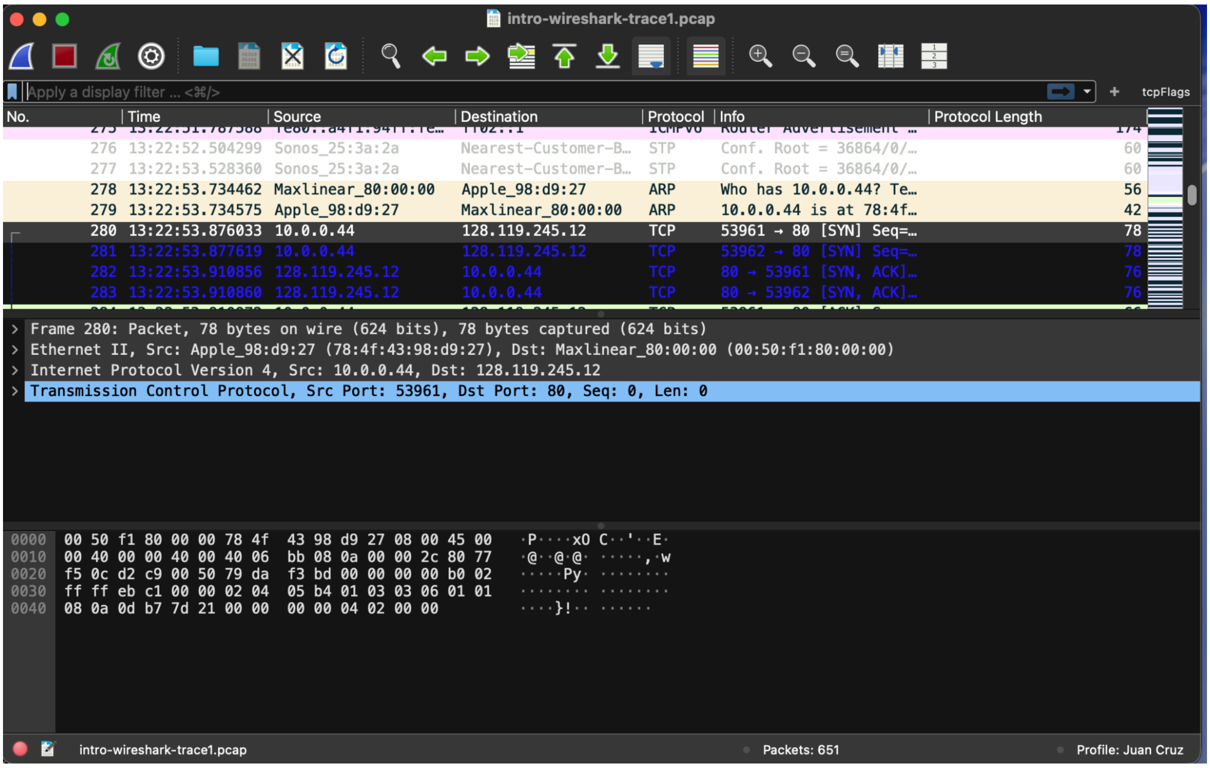
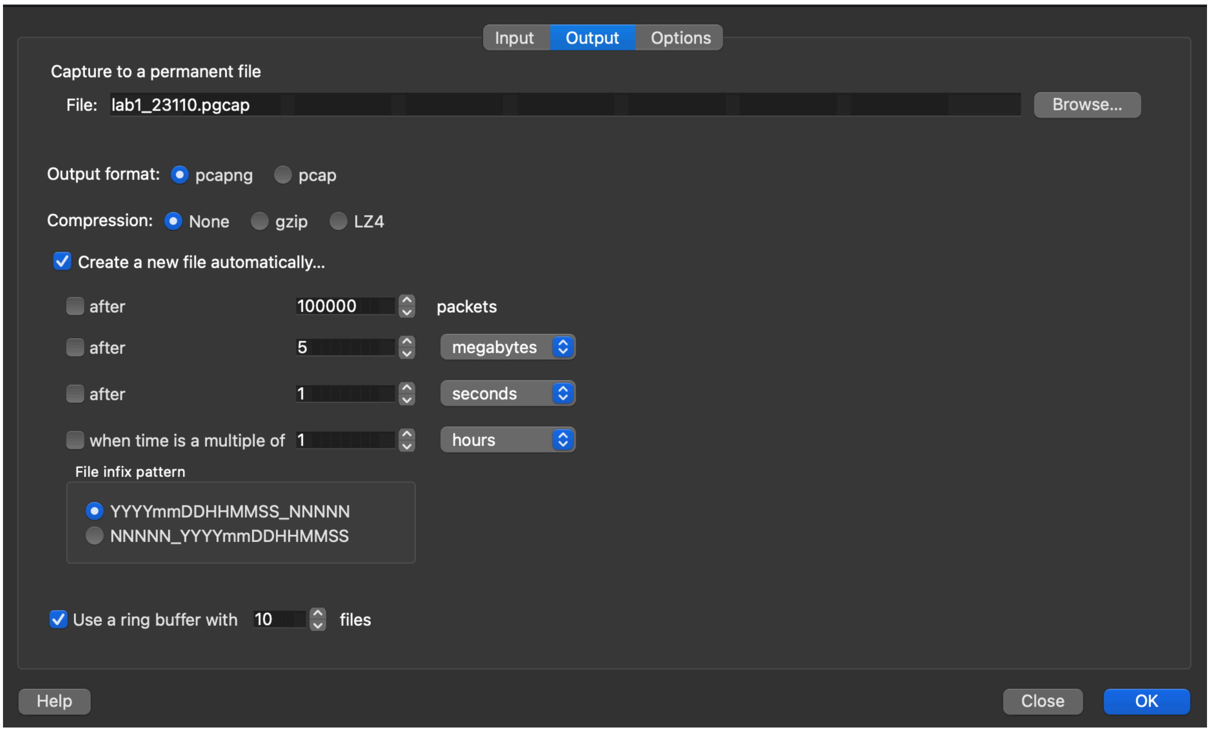
Guatemala 14 de Julio de 2026
Juan Cruz - 23110

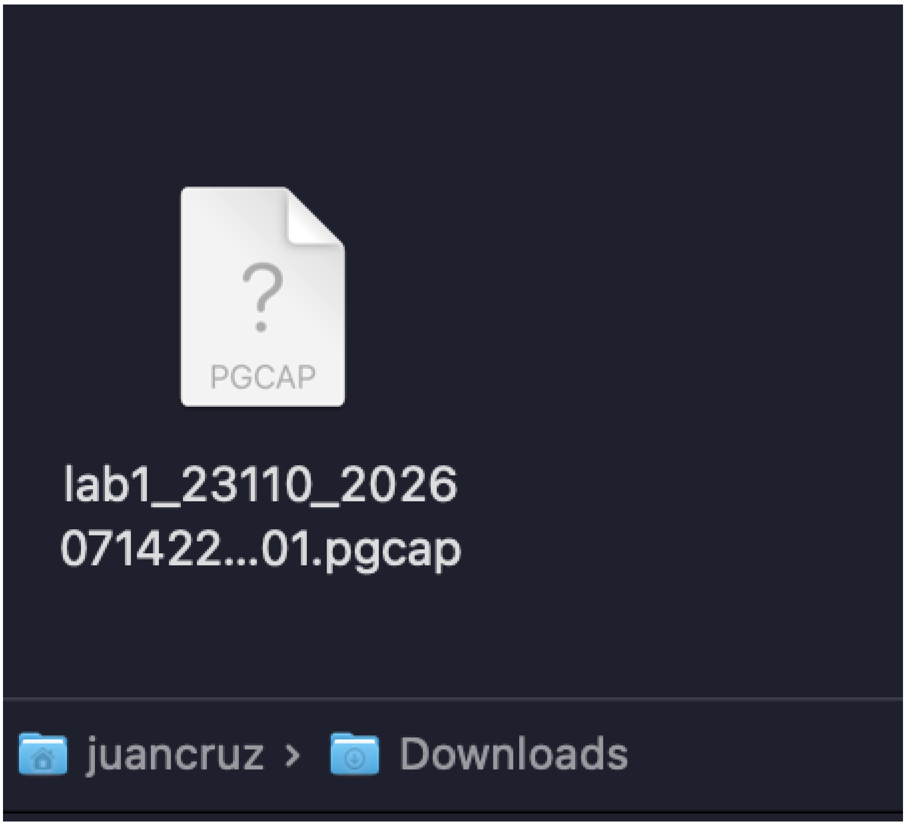
Redes

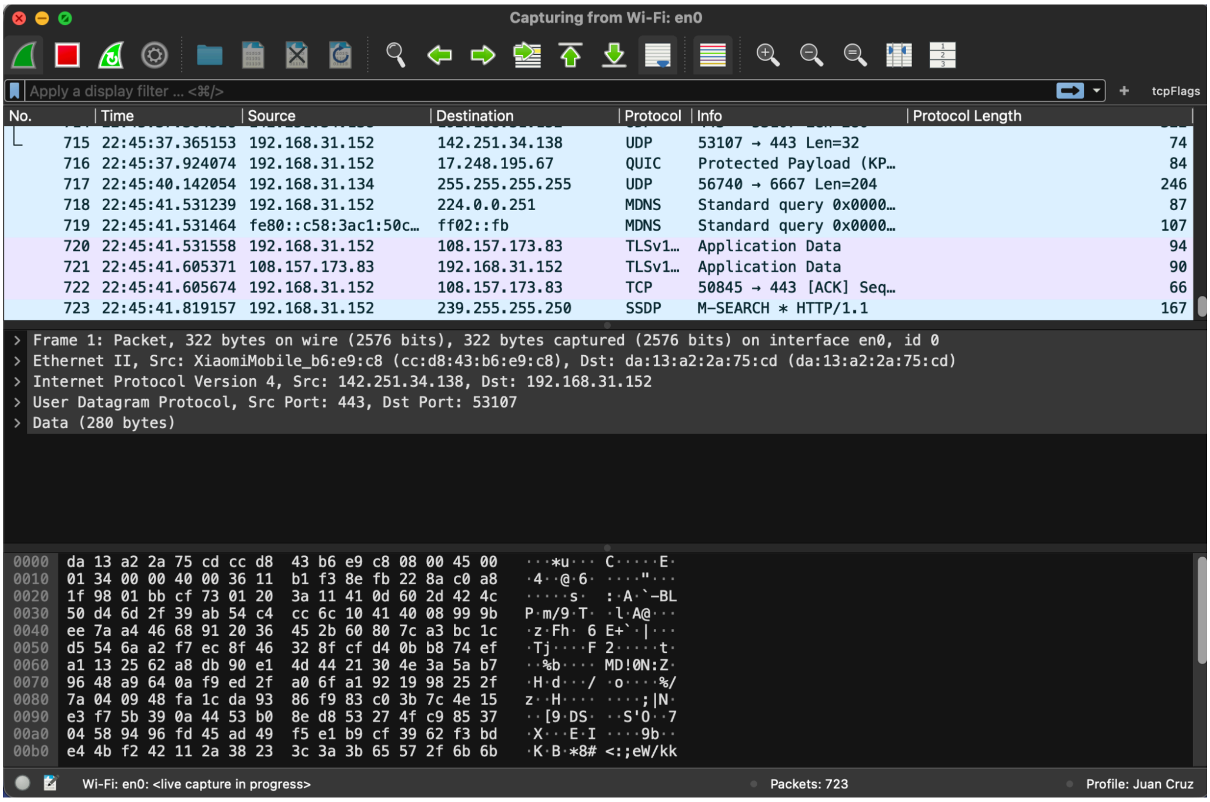
Laboratorio 1: Esquemas de comunicación

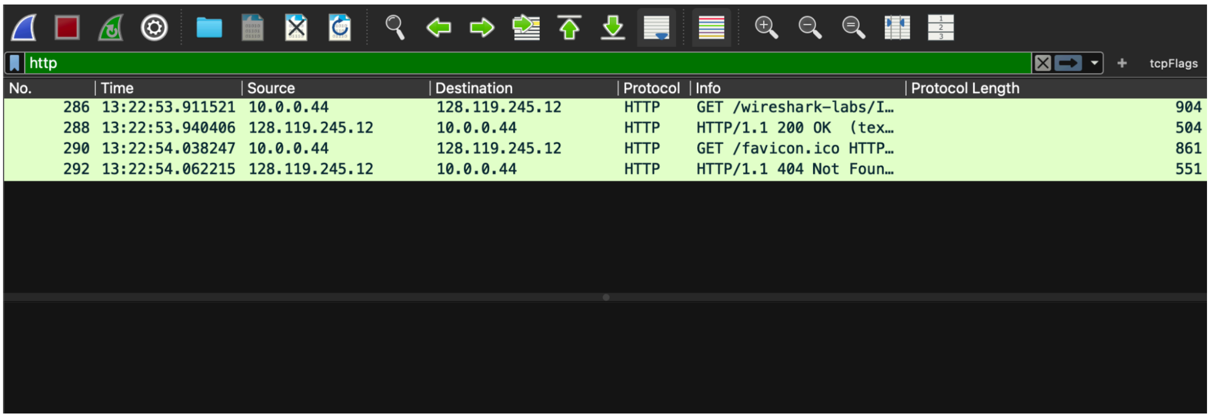


```
⌥⌘1  juancruz@Mac:~/Universidad
~/Universidad ............................................................ ✔ 10:29:05 PM
ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
        options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
        inet 127.0.0.1 netmask 0xff000000
        inet6 ::1 prefixlen 128
        inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
        nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether ee:3e:83:1c:97:c2
        media: none
        status: inactive
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether ee:3e:83:1c:97:c3
        media: none
        status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether ee:3e:83:1c:97:a2
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether ee:3e:83:1c:97:a3
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:02:a3:70:26:c0
        media: autoselect <full-duplex>
        status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:02:a3:70:26:c4
        media: autoselect <full-duplex>
        status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=63<RXCSUM,TXCSUM,TSO4,TSO6>
        ether 36:02:a3:70:26:c0
        Configuration:
                id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
                maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
                root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
                ipfilter disabled flags 0x0
        member: en1 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 8 priority 0 path cost 0
        member: en2 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 9 priority 0 path cost 0
        nd6 options=201<PERFORMNUD,DAD>
        media: <unknown type>
        status: inactive
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
        inet6 fe80::2cfd:539c:ac9e:a7cb%utun0 prefixlen 64 scopeid 0xf
```


Input
Output
Options
Capture to a permanent file
File: lab1_23110.pgcap
Browse...
Output format: pcapng pcap
Compression: None gzip LZ4
Create a new file automatically...
after 100000 packets
after 5 megabytes
after 1 seconds
when time is a multiple of 1 hours
File infix pattern
YYYYmmDDHHMMSS_NNNNN
NNNNN_YYYYmmDDHHMMSS
Use a ring buffer with 10 files
Help
Close
OK


PGCAP
lab1_23110_2026
071422...01.pgcap
juancruz › Downloads


Capturing from Wi-Fi: en0
Apply a display filter ... <⌘/>
tcpFlags
No. | Time | Source | Destination | Protocol | Info | Protocol Length
715 22:45:37.365153 192.168.31.152 142.251.34.138 UDP 53107 → 443 Len=32 74
716 22:45:37.924074 192.168.31.152 17.248.195.67 QUIC Protected Payload (KP… 84
717 22:45:40.142054 192.168.31.134 255.255.255.255 UDP 56740 → 6667 Len=204 246
718 22:45:41.531239 192.168.31.152 224.0.0.251 MDNS Standard query 0x0000… 87
719 22:45:41.531464 fe80::c58:3ac1:50c… ff02::fb MDNS Standard query 0x0000… 107
720 22:45:41.531558 192.168.31.152 108.157.173.83 TLSv1… Application Data 94
721 22:45:41.605371 108.157.173.83 192.168.31.152 TLSv1… Application Data 90
722 22:45:41.605674 192.168.31.152 108.157.173.83 TCP 50845 → 443 [ACK] Seq… 66
723 22:45:41.819157 192.168.31.152 239.255.255.250 SSDP M-SEARCH * HTTP/1.1 167
Frame 1: Packet, 322 bytes on wire (2576 bits), 322 bytes captured (2576 bits) on interface en0, id 0
Ethernet II, Src: XiaomiMobile_b6:e9:c8 (cc:d8:43:b6:e9:c8), Dst: da:13:a2:2a:75:cd (da:13:a2:2a:75:cd)
Internet Protocol Version 4, Src: 142.251.34.138, Dst: 192.168.31.152
User Datagram Protocol, Src Port: 443, Dst Port: 53107
Data (280 bytes)
0000 da 13 a2 2a 75 cd cc d8 43 b6 e9 c8 08 00 45 00
0010 01 34 00 00 40 00 36 11 b1 f3 8e fb 22 8a c0 a8
0020 1f 98 01 bb cf 73 01 20 3a 11 41 0d 60 2d 42 4c
0030 50 d4 6d 2f 39 ab 54 c4 cc 6c 10 41 40 08 99 9b
0040 ee 7a a4 46 68 91 20 36 45 2b 60 80 7c a3 bc 1c
0050 d5 54 6a a2 f7 ec 8f 46 32 8f cf d4 0b b8 74 ef
0060 a1 13 25 62 a8 db 90 e1 4d 44 21 30 4e 3a 5a b7
0070 96 48 a9 64 0a f9 ed 2f a0 6f a1 92 19 98 25 2f
0080 7a 04 09 48 fa 1c da 93 86 f9 83 c0 3b 7c 4e 15
0090 e3 f7 5b 39 0a 44 53 b0 8e d8 53 27 4f c9 85 37
00a0 04 58 94 96 fd 45 ad 49 f5 e1 b9 cf 39 62 f3 bd
00b0 e4 4b f2 42 11 2a 38 23 3c 3a 3b 65 57 2f 6b 6b
Wi-Fi: en0: <live capture in progress>
Packets: 723
Profile: Juan Cruz



Discusión:

En este laboratorio experimentamos las dos caras de la transmisión de datos. En la primera parte, comprobamos que los sistemas más simples (como el código Morse) generan menos errores al transmitirse. También vimos que incluir un conmutador es indispensable para ordenar los mensajes entre varios usuarios, aunque hace que la comunicación sea un poco más compleja.

En cuanto a la parte técnica, usar Wireshark fue un gran aprendizaje. Configurar el ring buffer me enseñó a hacer capturas de tráfico largas sin que la computadora se quede sin memoria. Además, al filtrar y analizar los paquetes HTTP, pude ver exactamente qué información se intercambia "por debajo" entre el navegador y el servidor al abrir una página web. En resumen, comprobé que este tipo de herramientas son esenciales para entender y arreglar problemas en cualquier red.